

Temporally Consistent RGB–CAD Recovery and Learned SE(3) Filtering for Long-Range Vehicle Pose Tracking

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Draft methods section for integration into a larger paper

Abstract

Per-frame category-level and novel-object pose estimators can produce accurate measurements under favorable viewing conditions, but their trajectories may contain depth outliers, abrupt translation changes, and approximately 180° front–rear orientation flips. We present a causal, sequence-aware post-processing pipeline for single-vehicle six-degree-of-freedom pose tracking. The method first expands each GigaPose estimate into RGB–CAD pose hypotheses, uses a gated recurrent unit (GRU) to select candidates over time, and resolves ambiguous orientation transitions with confidence-adaptive hysteresis and fixed-lag backfilling. A second learned error-state filter then predicts diagonal Kalman-style gains on SE(3) to smooth translation alone or the full pose. The design separates discrete hypothesis recovery from continuous motion filtering, resets all temporal state at sequence discontinuities, and retains measurement-relative safety fallbacks. On a 328-frame development-validation set containing distinct front- and rear-camera sequences, the fixed-lag selector reduced median translation and rotation error from 3.07 m and 15.50° to 0.66 m and 5.46°, respectively. A translation-only learned filter further reduced median translation error to 0.59 m while preserving the selector orientation exactly. These numbers are development-validation results; final claims should use the separate unseen benchmark described in Sec. 2.

1 Method

1.1 Problem formulation

For a monocular RGB sequence $\{I_t\}_{t=1}^T$, camera intrinsics K_t , an object mask or detection M_t , and a CAD model $\mathcal{C}$, the objective is to estimate the camera-to-object pose

$$T_t = T_{\text{camera,object},t} = \begin{bmatrix} R_t & \mathbf{p}_t \\ \mathbf{0}^\top & 1 \end{bmatrix} \in \text{SE}(3), \quad (1)$$

where $R_t \in \text{SO}(3)$ and $\mathbf{p}_t \in \mathbb{R}^3$. GigaPose [1] provides one or more per-frame measurements $\{T_{t,k}^{\text{GP}}\}$. Although these estimates contain useful appearance and geometric evidence, per-frame inference does not enforce trajectory consistency. Our goal is therefore to recover a temporally coherent pose without requiring depth, LiDAR, or a map at inference time.

We assume one target vehicle per sequence. Consequently, the present method does not solve multi-object association; its temporal state is reset whenever the source run, camera, or continuous frame segment changes.

1.2 RGB–CAD hypothesis generation

For each frame, we construct a bounded candidate set $\mathcal{H}_t = \{T_{t,j}\}_{j=1}^N$ around the top GigaPose hypotheses. The set includes translation/depth offsets, image-center offsets, local rotation perturbations, explicit yaw perturbations, and an approximately 180° orientation alternative. A matching network combines spatial RGB/mask features, rendered CAD features, and frozen DINOv2 patch features [2]. Each hypothesis is represented by a feature vector

$$\mathbf{c}_{t,j} = [q_{t,j}, s_{t,j}, \text{IoU}_{t,j}, \Delta u_{t,j}, \Delta \log z_{t,j}, \Delta R_{t,j}, \Delta T_{t,j}^{\text{GP}}, \mathbf{r}_{t,j}^{6\text{D}}, \mathbf{b}_{t,j}], \quad (2)$$

where q is confidence, s is the learned RGB–CAD quality score, IoU measures rendered-mask agreement, $\mathbf{r}^{6\text{D}}$ is a continuous rotation representation, and $\mathbf{b}$ encodes the proposal type.

The candidate exporter intentionally assigns zero weight to its legacy history term. This prevents a hand-designed temporal score from eliminating a valid recovery hypothesis before the learned sequence model observes it. All temporal selection is delegated to the GRU and orientation resolver.

1.3 Causal GRU candidate selection

Let $\mathbf{f}_t$ denote frame-level measurement and detection context. A causal GRU updates its hidden state and predicts a logit for each candidate:

$$\mathbf{h}_t = \text{GRU}(\{\mathbf{c}_{t,j}\}_{j=1}^N, \mathbf{f}_t, \mathbf{h}_{t-1}), \quad (3)$$

$$\ell_{t,j} = g_\theta(\mathbf{c}_{t,j}, \mathbf{f}_t, \mathbf{h}_t), \quad (4)$$

$$\pi_{t,j} = \frac{\exp(\ell_{t,j})}{\sum_k \exp(\ell_{t,k})}. \quad (5)$$

The selector is trained with cross-entropy against the candidate having the lowest normalized pose error. Translation and geodesic rotation errors are combined only to construct this oracle candidate label; ground truth is not used during inference.

Because a selector cannot choose a pose that is absent from $\mathcal{H}_t$, the proposal set includes orientations based on

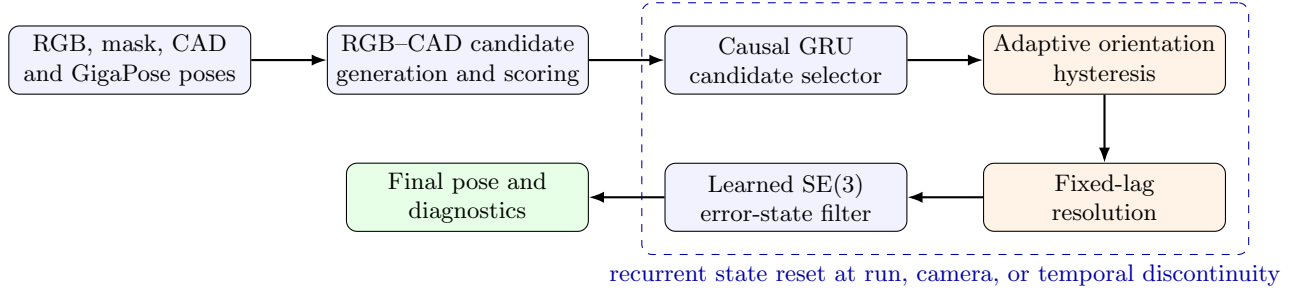


Figure 1: Inference pipeline. The candidate selector addresses multimodal and discrete pose ambiguity, especially front–rear flips. The learned error-state filter subsequently performs continuous translation or full-pose smoothing. Adaptive orientation and fixed-lag decisions occur before the learned filter.

the previous accepted pose and constant angular velocity, in addition to perturbations of the current GigaPose pose. This is essential when the current per-frame estimate is already front–rear flipped.

1.4 Adaptive orientation hysteresis

The temporal orientation prediction is

$$\hat{R}_t^- = R_{t-1} \text{Exp}(\omega_{t-1} \Delta t), \quad (6)$$

and candidate j is compared to it with the geodesic distance

$$\theta_{t,j} = \left\| \text{Log} \left((\hat{R}_t^-)^\top R_{t,j} \right) \right\|_2. \quad (7)$$

Candidates below 45° receive no additional penalty; candidates between 45° and 90° receive a soft rotation-jump penalty. Changes above 135° are treated as transitions between persistent front–rear modes, rather than ordinary orientation updates.

Let $m_t \in \{0, 1\}$ denote the persistent orientation mode. A proposed mode change opens a pending interval. The required confirmation length adapts to the stability of the current mode:

$$n_{\text{confirm}} = \begin{cases} 2, & n_{\text{stable}} < 4, \\ 5, & n_{\text{stable}} \geq 4. \end{cases} \quad (8)$$

Pending observations must be mutually consistent within 45° and meet minimum selector probability and margin requirements. A normal candidate consistent with the established mode cancels the pending transition. Thus, an isolated high-confidence error cannot immediately reverse a stable track.

1.5 Fixed-lag orientation resolution

Causal hysteresis alone protects the current output by retaining its temporal orientation during an unresolved transition. If the transition later proves genuine, however, those earlier frames would remain assigned to the old mode. We therefore buffer the pending interval. Once the evidence confirms a mode change, the resolver backfills the interval with the selected orientations in the confirmed mode. If the

transition is rejected, the buffered interval is committed in the previous mode. The maximum delay equals the adaptive confirmation horizon (two or five frames in our implementation).

This operation is a fixed-lag smoother only for the discrete orientation mode; it does not use future information beyond the bounded pending interval and does not alter translation.

1.6 Learned SE(3) error-state filtering

After fixed-lag resolution, a KalmanNet-style recurrent filter [3] performs continuous smoothing. Its state is

$$\mathbf{x}_t = (\mathbf{p}_t, R_t, \mathbf{v}_t, \omega_t), \quad (9)$$

with constant linear- and angular-velocity prediction

$$\hat{\mathbf{p}}_t = \mathbf{p}_{t-1} + \mathbf{v}_{t-1} \Delta t, \quad (10)$$

$$\hat{R}_t^- = R_{t-1} \text{Exp}(\omega_{t-1} \Delta t). \quad (11)$$

Given measurement $(\mathbf{p}_t^m, R_t^m)$, the innovations are

$$\mathbf{e}_t^p = \mathbf{p}_t^m - \hat{\mathbf{p}}_t, \quad (12)$$

$$\mathbf{e}_t^R = \text{Log} \left((\hat{R}_t^-)^\top R_t^m \right). \quad (13)$$

The recurrent input concatenates normalized innovations, the previous linear and angular velocities, Δt , and measurement context

$$\mathbf{u}_t = [\mathbf{e}_t^p, \mathbf{e}_t^R, \mathbf{v}_{t-1}, \omega_{t-1}, \Delta t, \mathbf{d}_t], \quad (14)$$

where $\mathbf{d}_t$ contains measurement confidence, normalized bounding-box center, width, height, area, and measured depth. A GRU predicts 12 diagonal gains in $(0, 1)$:

$$[\mathbf{k}_t^p, \mathbf{k}_t^R, \mathbf{k}_t^v, \mathbf{k}_t^\omega] = \sigma(g_\phi(\text{GRU}(\mathbf{u}_t))). \quad (15)$$

The update is

$$\mathbf{p}_t = \hat{\mathbf{p}}_t + \mathbf{k}_t^p \odot \mathbf{e}_t^p, \quad (16)$$

$$R_t = \hat{R}_t^- \text{Exp}(\mathbf{k}_t^R \odot \mathbf{e}_t^R), \quad (17)$$

$$\mathbf{v}_t = \mathbf{v}_{t-1} + \mathbf{k}_t^v \odot \mathbf{e}_t^p / \Delta t, \quad (18)$$

$$\omega_t = \omega_{t-1} + \mathbf{k}_t^\omega \odot \mathbf{e}_t^R / \Delta t. \quad (19)$$

Unlike a classical Kalman filter, the method does not explicitly propagate covariance matrices P , Q , and R . The learned gains approximate an adaptive correction policy while the explicit SE(3) prediction preserves the state-space structure.

We train two variants. The *translation-only* model applies the learned translation update but copies the fixed-lag selector rotation exactly:

$$R_t^{\text{out}} = R_t^m. \quad (20)$$

The *full-pose* model applies both learned translation and rotation updates. At inference, corrections exceeding configured translation or rotation limits are rejected in favor of the input measurement.

1.7 Filter training objective

For valid frames after the first element of each clip, the full-pose filter is trained using

$$\mathcal{L}_{\text{pose}} = \frac{\|\mathbf{p}_t - \mathbf{p}_t^*\|_2}{s_p} + \frac{d_{\text{SO}(3)}(R_t, R_t^*)}{s_R}, \quad (21)$$

where

$$d_{\text{SO}(3)}(R_1, R_2) = \|\text{Log}(R_1^\top R_2)\|_2. \quad (22)$$

A weak measurement-preservation regularizer discourages unnecessary changes:

$$\mathcal{L} = \mathcal{L}_{\text{pose}} + \lambda_{\text{pres}} \left(\frac{\|\mathbf{p}_t - \mathbf{p}_t^m\|_2}{s_p} + \frac{d_{\text{SO}(3)}(R_t, R_t^m)}{s_R} \right). \quad (23)$$

For the translation-only model, both rotation terms are omitted and the input rotation is preserved by construction. Models are trained on consecutive clips; clips never cross run, camera, or discontinuity boundaries.

1.8 Relation to GRUTrack

The proposed filter shares the learned-gain motivation of KalmanNet and the open-source GRUTrack system [4]. GRUTrack is a complete nuScenes multi-object tracker that combines learned Kalman gains with data association, track lifecycle management, and automotive box states such as position, dimensions, planar velocity, acceleration, yaw, and yaw rate. In contrast, our model assumes one already-associated vehicle, operates directly on camera-to-object poses in SE(3), and estimates full three-dimensional rotation and angular velocity. The RGB-CAD candidate selector and fixed-lag front-rear state are also specific to our formulation. Accordingly, the implementation should be viewed as a task-specific KalmanNet-style filter, not as a reproduction of GRUTrack.

2 Experimental Protocol

Data separation. Assetto Corsa sequences are partitioned by physical source run, not by random frame. Consecutive clips from one run therefore cannot appear in

both training and evaluation. Front and rear cameras are represented separately, and all recurrent state is reset when the source run, camera, frame continuity, or timestamp continuity changes. A final benchmark contains 5,174 usable frames from four unseen sequences after excluding 361 manifest rows without an unambiguous joined 3D box/instance-mask identifier. It covers distances from 0–140 m. Since training used samples only through 120 m, the 120–140 m bin must be reported separately as out-of-distribution.

Metrics. Translation error and geodesic rotation error are

$$e_t = \|\mathbf{p}_t - \mathbf{p}_t^*\|_2, \quad (24)$$

$$e_R = \frac{180}{\pi} \cos^{-1} \left(\text{clip} \left(\frac{\text{tr}(R_t^\top R_t^*) - 1}{2}, -1, 1 \right) \right). \quad (25)$$

We report mean, median, and 90th percentile errors; recall below 1 m and 10°; per-distance-bin performance; and paired frame-level improvement fractions relative to GigaPose. Front and rear cameras should additionally be reported separately.

Required ablations. The final comparison should include: (i) GigaPose; (ii) standalone RGB self-recovery; (iii) GRU candidate selection without adaptive orientation; (iv) adaptive orientation without fixed lag; (v) the complete fixed-lag selector; (vi) a GRU-Kalman filter applied directly to raw GigaPose; and (vii) fixed-lag selector followed by translation-only or full-pose learned filtering. Candidate generation and evaluation frames must be identical across these conditions.

3 Development-Validation Results

Table 1 summarizes the currently available 328-frame development-validation result containing Laguna front and Putnam rear sequences. Checkpoints were selected using this development regime; therefore, these values demonstrate pipeline behavior but must not be presented as final unseen-test performance.

Relative to GigaPose, the fixed-lag selector improved translation on 85.1% of paired frames and rotation on 67.4%. The translation-only cascade improved translation on 88.1% of frames while retaining the same orientation as the selector. Full-pose filtering improved rotation on 77.4% of paired frames, but its 10° recall was slightly lower than that of the selector. This trade-off motivates reporting both cascade variants rather than selecting one solely from an aggregate loss.

4 Limitations and Scope

The current formulation assumes one known CAD object and does not perform multi-object data association. The

Table 1: Development-validation results on 328 frames. Lower error is better; higher recall is better. “T-only” preserves fixed-lag rotation exactly.

Method	Med. e_t (m)	P90 e_t (m)	Med. e_R ($^\circ$)	P90 e_R ($^\circ$)	$e_t \leq 1\text{m}$	$e_R \leq 10^\circ$
GigaPose	3.072	17.786	15.496	163.026	10.98%	41.16%
GRU selector + adaptive fixed lag	0.662	7.697	5.464	23.035	58.84%	70.43%
+ learned filter (T-only)	0.594	5.540	5.464	23.035	61.89%	70.43%
+ learned filter (full pose)	0.592	6.057	4.728	23.238	62.20%	69.51%

learned filter cannot create a correct hypothesis when every upstream candidate is wrong; recovery from catastrophic measurements is primarily the responsibility of candidate generation and the selector. Fixed-lag hysteresis adds a bounded output delay. Its thresholds are currently fixed hyperparameters and may require calibration under different frame rates. The simulator-to-real gap remains a central risk: appearance, mask quality, motion statistics, camera calibration, and GigaPose error modes can all shift. Finally, the reported development-validation values should not be interpreted as final generalization estimates. The unseen 5,174-frame benchmark and a real-world sequence evaluation are required before making conclusive performance claims.

Reproducibility note. For every experiment, the paper should record the GigaPose checkpoint, RGB-CAD checkpoint, GRU selector checkpoint, learned-filter checkpoint, candidate configuration, sequence-gap thresholds, orientation-hysteresis thresholds, fixed-lag setting, correction guards, translation units, CAD origin, camera split, and exact physical run identifiers.

References

- [1] Van Nguyen Nguyen, Thibault Groueix, Mathieu Salzmann, and Vincent Lepetit. Gigapose: Fast and robust novel object pose estimation via one correspondence. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [2] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- [3] Guy Revach, Nir Shlezinger, Xiaoyong Ni, Adria Lopez Escoriza, Ruud J. G. van Sloun, and Yonina C. Eldar. KalmanNet: Neural network aided kalman filtering for partially known dynamics. *IEEE Transactions on Signal Processing*, 70:1532–1547, 2022.
- [4] Xiang-1208. GRUTrack: Neural-network-based kalman filtering for 3d multi-object tracking. <https://github.com/xiang-1208/GRUTrack>, 2025. Accessed 2026-08-04.